

Homework Grading Report

Student:	Student_10
Assignment:	Midterm Exam (mt13)
Date:	February 08, 2026
Grade:	119.3 / 125 (95%) A

Section Breakdown

Section	Points	Earned	Notes
Part 1: R Basics and Data Import	10	10.0	All tasks completed
Part 2: Data Cleaning	15	7.5	Missing: drop_na(), IQR()
Part 3: Basic Data Transformation	15	15.0	Missing: %>%()
Part 4: Advanced Transformation	15	15.0	All tasks completed
Part 5: Data Reshaping	15	15.0	All tasks completed
Part 6: Combining Datasets	15	15.0	All tasks completed
Part 7: String Manipulation & Date/Time	20	20.0	All tasks completed
Part 8: Advanced Wrangling	15	15.0	All tasks completed
Part 9: Reflection Questions	5	5.0	All tasks completed
TOTAL	125	119.3	95%

Detailed Feedback

What You Did Well

- Part 1: R Basics and Data Import (Lesson 1) - Task 1.1: Set Working Directory
- Part 1: R Basics and Data Import (Lesson 1) - Task 1.2: Load Required Packages
- Part 1: R Basics and Data Import (Lesson 1) - Task 1.3: Import Datasets
- Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.2: Handle Missing Values
- Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.3: Detect Outliers in Revenue
- Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.3: Sort by Revenue
- Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.4: Chain Multiple Operations
- Part 4: Data Transformation Part 2 (Lesson 4) - Task 4.2: Calculate Overall Summary Statistics

What to Fix

- 1. To strengthen your data:** quality verification, explicitly run functions like `any(is.na())` and `boxplot()` to detect hidden missing values or outliers; this provides evidence for your claim that none exist. The solution example uses `sum(is.na(sales_data$Revenue))` and `ggplot2::geom_boxplot()` to surface issues.
- 2.** When discussing grouped insights, quantify the findings (e.g., "North region generated \$1.2M in sales, 15% above the company average"). Quantitative statements allow stakeholders to gauge magnitude and prioritize actions. The reference solution reports exact sales totals and percentage differences.
- 3. Avoid repeating the same example in your write:** vs-long explanation; instead, present two distinct business cases, such as a dashboard report (wide) and a churn-prediction model (long). The model solution contrasts a quarterly sales matrix with a transaction-level log.
- 4.** In your join discussion, demonstrate how to validate join results by checking for unmatched keys with `anti_join()` and reporting the count of orphan records. The exemplar code uses `anti_join(customers, orders, by = "CustomerID") %>% tally()`.
- 5.** For the KPI argument, cite a specific metric you created (e.g., "Average Order Value = $\text{Total_Amount} / \text{Number_of_Orders}$ ") and explain how it informs profitability assessments. The solution showcases a KPI table with columns for `Revenue_per_Customer` and `Return_Rate`.

Reflection & Critical Thinking

In 9.1 you correctly note that the dataset contained no missing values or outliers, and you explain why cleaning would matter for reliability and bias. However, you did not illustrate how you verified the absence of issues (e.g., using `summary()` or visual checks), which would demonstrate a more rigorous approach.

For 9.2 you identify that grouped analysis reveals regional and category performance differences and link this to marketing and inventory decisions. To deepen the insight, you could reference specific metrics you calculated (e.g., total sales by region) and explain how those numbers drive strategic choices.

Your answer to 9.3 accurately describes when wide and long formats are useful and provides business scenarios for each. The response repeats the same example twice, reducing clarity; a distinct second scenario (e.g., long format for time-series forecasting) would show broader understanding.

In 9.4 you correctly differentiate `left_join()` and `inner_join()` and give appropriate business contexts for each. Adding a brief note on potential data loss with `inner_join()` and how to check join integrity would strengthen the explanation.

For 9.5 you argue that Lesson 8 is most valuable because it creates categories and KPIs, linking these to strategic decisions. Including a concrete KPI you built (e.g., profit margin per category) would illustrate the practical impact of that skill.

Instructor Assessment

You have built a solid technical foundation by importing the files, performing joins, and using tidyverse verbs correctly. Your reflections show awareness of why cleaning, grouping, and reshaping matter, but they would be stronger with concrete numbers and clearer differentiation of examples. Focus on quantifying your insights and explicitly documenting validation steps to elevate both analytical rigor and business relevance.

Key Takeaway

Complete a brief data-quality audit for each dataset and document the steps you took to confirm completeness.

When describing insights, always pair qualitative observations with numeric evidence to make your conclusions actionable.

Practice writing concise yet distinct examples for conceptual questions to avoid redundancy and demonstrate breadth of understanding.